

$$c_1 = \sqrt{l_1} = 1$$

12. **Ans: A**

The rate of heat loss of the hot liquid is constant during cooling before solidification and during the freezing

During cooling:

$$Pt = mc\Delta\theta$$

$$P(1) = mc(4.0)$$

During freezing:

$$Pt = ml_f$$

$$P(25) = ml_f$$

$$\frac{P(1)}{P(25)} = \frac{mc(4.0)}{ml_f}$$

$$\frac{c}{l_f} = \frac{1}{25} \times \frac{1}{4.0} = 0.01$$

13. **Ans: B**